Zhiyang He (Sunny)

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Education

Massachusetts Institute of Technology 2021 - 2027

Ph.D. in Applied Math and Theoretical Computer Science Advisors: Peter Shor, Anand Natarajan

Carnegie Mellon University 2016 - 2021

B.S. & M.S. in Mathematical Science, B.S. in Computer Science Advisor: Boris Bukh

Research Focus

My research studies and designs fault-tolerant quantum computing systems, with speciality in quantum LDPC codes and code surgery. I am broadly interested in quantum algorithms and complexity theory.

Recent Works

Extractor Architectures: A theoretical blueprint for large-scale, LDPC-based fault-tolerant quantum computers. Presented at QEC 2025 and QIP 2026. Received the Charles W. and Jennifer C. Johnson Prize from MIT Mathematics in 2026, which is awarded to one paper in applied math each year. [arXiv]

Experimental Demonstration of Logical Magic State Distillation: An experimental demonstration of magic state distillation with logical qubits encoded in color codes, performed on QuEra's Gemini-class neutral atom computer. Presented at QEC 2025. Published in Nature. [Nature]

Logical Processing on Bivariate Bicycle Code: Improved theoretical techniques of code surgery which, when applied to the 144-qubit bivariate bicycle code, enable full logical processing of the 12 logical qubits. Presented at QIP 2025. [arXiv]

Research, Teaching and Services

Program Committee Services: QIP 2026, TQC 2026

Workshop organization: FOCS 2026 workshop on Quantum Architectures.

Student Researcher, Google Quantum AI Summer 2026

Quantum Error Correction Intern, QuEra Computing Summer 2024

Quantum Research Intern, IBM Research Summer 2022, 2023

Teaching Assistant, Massachusetts Institute of Technology

- TA | Linear Algebra (18-06) Feb 2026 - May 2026
- TA | Quantum Computation (18-435) Sep 2022 - Dec 2022

Teaching Assistant & Grader, Carnegie Mellon University

- TA | Discrete Mathematics (21-228) Jan 2020 - May 2020
- TA | Integration and Approximations (21-122) Jan 2019 - May 2019
- TA | Concepts of Mathematics (21-128) Jan 2018 - May 2018

Fellowships and Awards

Charles W. and Jennifer C. Johnson Prize 2026

MIT-IBM Watson AI Fellowship 2025-2027

NSF Graduate Research Fellowship 2023-2027

MIT Presidential Fellowship 2021-2022

CMU Math Honors 2021

Publication and Preprints

1. **Efficient Logic with Ultra-High-Rate Quantum Codes.**
   Nishad Maskara, Rohan Mehta, Zhiyang He, Varun Menon, J. Pablo Bonilla Ataides, Mikhail D. Lukin, Hengyun Zhou. [arXiv]
2. **Low-Overhead Quantum Error Correction with Boundary-Connected Planar Modules.**
   Oscar Higgott, Hasan Sayginel, Francisco J.H. Heras, Zhiyang He, Tomas Jochym-O'Connor, Andrew W. Senior, Lei M. Zhang, Thomas Edlich, James S. Spencer, Matt McEwen, Craig Gidney, Johannes Bausch, Pushmeet Kohli, Hartmut Neven.
   QEC 2026. [arXiv]
3. **Full Extractors for Logical Processing in Hypergraph Product Codes.**
   John Blue, Zhiyang He, Hengyun Zhou, Isaac L. Chuang. [arXiv]
4. **In-Situ Simultaneous Magic State Injection on Arbitrary CSS qLDPC Codes.**
   Kun Liu, Shifan Xu, Tomas Jochym-O'Connor, Zhiyang He, Shraddha Singh, Yongshan Ding. [arXiv]
5. **Parsimonious Quantum Low-Density Parity-Check Code Surgery.**
   Andrew C. Yuan, Alexander Cowtan, Zhiyang He, Ting-Chun Lin, Dominic J. Williamson. [arXiv]
6. **Constant-Time Surgery on 2D Hypergraph Product Codes with Near-Constant Space Overhead.**
   Kathleen Chang, Zhiyang He, Theodore J. Yoder, Guanyu Zhu, Tomas Jochym-O'Connor.
   QEC 2026. [arXiv]
7. **Distilling Magic States in the Bicycle Architecture.**
   Shifan Xu, Kun Liu, Patrick J. Rall, Zhiyang He, Yongshan Ding.
   International Symposium on Computer Architecture (ISCA) 2026. [arXiv]
8. **Fast and fault-tolerant logical measurements: Auxiliary hypergraphs and transversal surgery.**
   Alexander Cowtan, Zhiyang He, Dominic J. Williamson, Theodore J. Yoder.
   QEC 2026. [arXiv]
9. **Layer codes as partially self-correcting quantum memories.**
   Shouzhen Gu, Libor Caha, Shin Ho Choe, Zhiyang He, Aleksander Kubica, Eugene Tang.
   QIP 2026, Published in *Physics Review Letters.* [arXiv]
10. **Characterization of permutation gates in the third level of the Clifford hierarchy.**
    Zhiyang He, Luke Robitaille, Xinyu Tan.
    TQC 2026. [arXiv]
11. **Composable Quantum Fault-Tolerance.**
    Zhiyang He, Quynh T. Nguyen, Christopher A. Pattison.
    QEC 2025. [arXiv]
12. **Machine Learning Decoding of Circuit-Level Noise for Bivariate Bicycle Codes.**
    John Blue, Harshil Avlani, Zhiyang He, Liu Ziyin, Isaac L. Chuang.
    Published in *Quantum.* [arXiv]
13. **Extractors: QLDPC Architecture for Efficient Pauli-Based Computation.**
    Zhiyang He, Alexander Cowtan, Dominic J. Williamson, Theodore J. Yoder.
    QEC 2025, QIP 2026. [arXiv]
14. **Parallel Logical Measurements via Quantum Code Surgery.**
    Alexander Cowtan, Zhiyang He, Dominic J. Williamson, Theodore J. Yoder.
    QEC 2025, TQC 2025. Published in *Physics Review X Quantum*. [arXiv]
15. **(a) Quantum Codes with Addressable and Transversal Non-Clifford Gates.**
    (b) Asymptotically Good Quantum Codes with Addressable and Transversal Non-Clifford Gates.
    Zhiyang He, Vinod Vaikuntanathan, Adam Wills, Rachel Yun Zhang.
    QEC 2025, QIP 2026. [arXiv (a)], [arXiv (b)]

16. **Experimental Demonstration of Logical Magic State Distillation.**
QuEra Team and Collaborators. Contributed as lead theorist for experimental design.
Published in *Nature (2025)*. [arXiv]
17. **Improved QLDPC Surgery: Logical Measurements and Bridging Codes.**
Andrew Cross, Zhiyang He, Patrick Rall, Theodore Yoder.
QIP 2025. Published in *Nature Partner Journals Quantum Information.* [arXiv]
18. **Single-shot decoding of good quantum LDPC codes.**
Shouzhen Gu, Eugene Tang, Libor Caha, Shin Ho Choe, Zhiyang He, Aleksander Kubica.
QEC 2023, TQC2024. Published in *Communications in Mathematical Physics.* [arXiv]
19. **Quantum Locally Testable Codes with Constant Soundness.**
Andrew Cross, Zhiyang He, Anand Natarajan, Mario Szegedy, Guanyu Zhu.
QIP 2023. Published in *Quantum.* [arXiv]
20. **Breaking the n^k Barrier for Minimum k-cut on Simple Graphs**.
Zhiyang He, Jason Li.
STOC 2022. [arXiv]
21. **Near-linear-time, Optimal Vertex Cut Sparsifier in Directed Acyclic Graphs**.
Zhiyang He, Jason Li, Magnus Wahlström.
ESA 2021, **Best Paper Award**. [arXiv]
22. **New Upper Bound on Extremal Number of Even Cycles**.
Zhiyang He.
The Electronic Journal of Combinatorics (2021): P2-41. [arXiv]
23. **Hypergraphs with Few Berge Paths of Fixed Length between Vertices**.
Zhiyang He, Michael Tait.
SIAM Journal on Discrete Mathematics 33.3 (2019): 1472-1481. [arXiv]

Invited Talks & Presentations

1. **A Tutorial on QLDPC Surgery** [Link]. Presented at:
   - Microsoft Quantum Seminar, Nov 2026 [scheduled]
   - YITP Quantum Error Correction Workshop, Sep 2026
   - Benasque FTQT Workshop, Aug 2026
   - Google Quantum AI Seminar, Aug 2026
   - MIT Quantum Error Correction Seminar, Mar 2026
2. **Building a Quantum Computer with QLDPC Codes** [Link]. Presented at:
   - FOCS 2026 Workshop on Quantum Architectures, Nov 2026 [scheduled]
   - Simons Institute Workshop, Jul 2026
   - MIT Complexity Seminar, Mar 2025
3. **Extractors for Modular Hyperbolic Memories**. Presented at:
   - Google Quantum AI Seminar, Aug 2026
4. **Extractors: QLDPC Architecture for Efficient Pauli-Based Computation** [Link]. Presented at:
   - QIP 2026, Jan 2026
   - IEEE Quantum Week, Sep 2025
   - QEC 2025, Aug 2025
   - Simons Institute Workshop, May 2025
   - Google Quantum AI Seminar, Jun 2026

- University of Edinburgh Quantum Seminar, Jan 2026
- Tufts University Quantum Seminar, Nov 2025
- Xanadu Quantum Seminar, May 2025
- Rutgers DIMACS Workshop, May 2025
- Yale Quantum Initiative Symposium, May 2025
- MIT Quantum Information Seminar, Apr 2025
- University of Sydney and PsiQuantum Seminar, Apr 2025

5. **Composable Quantum Fault Tolerance** [Link]. Presented at:
   - Foxconn Quantum Research Seminar, Nov 2025
6. **Challenges in Scalable Quantum Error Correction** [Link]. Presented at:
   - IBM Mathematics of Computation and Algorithms Workshop, Oct 2025
7. **Constructions of Surgery Ancilla Systems: A Reflection on Ideas and Challenges**. Presented at:
   - IBM Thomas J. Watson Research Center, Aug 2025
8. **Quantum Codes with Addressable and Transversal Non-Clifford Gates** [Link]. Presented at:
   - Inria Quantum Architecture Seminar, May 2025
9. **Improved QLDPC Surgery: Logical Measurements and Bridging Codes** [Link]. Presented at:
   - QIP 2025, Feb 2025
   - MIT Quantum Information Seminar, Feb 2025
   - Harvard Quantum Computing Seminar, Feb 2025
   - Benasque FTQT Workshop, Aug 2024
10. **Introduction to Quantum Error Correction**. Presented at:
    - MIT Quantum Fault Tolerance Seminar, Sep 2024
    - CMU Quantum Error Correction Seminar, Feb 2020
11. **Single-shot decoding of good quantum LDPC codes** [Link]. Presented at:
    - TQC 2024, Sep 2024
12. **An Introduction to Quantum LDPC Codes** [Link]. Presented at:
    - UCLA Guest Lecture, Mar 2024
13. **Quantum LTCs with Constant Soundness** [Link]. Presented at:
    - QIP 2023, Feb 2023
    - MIT Quantum Information Seminar, Nov 2022
    - Tufts University Quantum Complexity Seminar, Oct 2022
    - IBM Thomas J. Watson Research Center, Sep 2022
14. **Breaking the n^k Barrier for Minimum k-cut on Simple Graphs**. Presented at:
    - STOC 2022, Jun 2022
    - MIT TCS Seminar, Nov 2021
15. **Asymptotically Good QLDPC Codes**. Presented at:
    - MIT Quantum LDPC Codes Seminar, Mar 2022
16. **Near-linear-time, Optimal Vertex Cut Sparsifier in Directed Acyclic Graphs**. Presented at:
    - ESA 2021, Sep 2021. **Best Paper Talk.**

17. **A New Upper Bound on Extremal Number of Even Cycles**. Presented at:
    - CMU Combinatorics Seminar, Mar 2019